SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1678 **COURSE NAME: Data Warehousing & Data Mining for Cluster Analysis**

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks: 50**

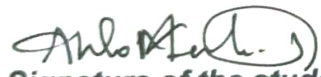
Annexure A

SHORT ANSWER TEST **Duration :**

Code of Conduct:

I Anlo Melwin S-192511299 ***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.***

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total (50) Marks	Out of 100
Weightage	30%	25%	20%	15%	10%	100%	
Q1							
Q2							
Q3							
Q4							
Q5							

RUBRICS FOR CALCULATION

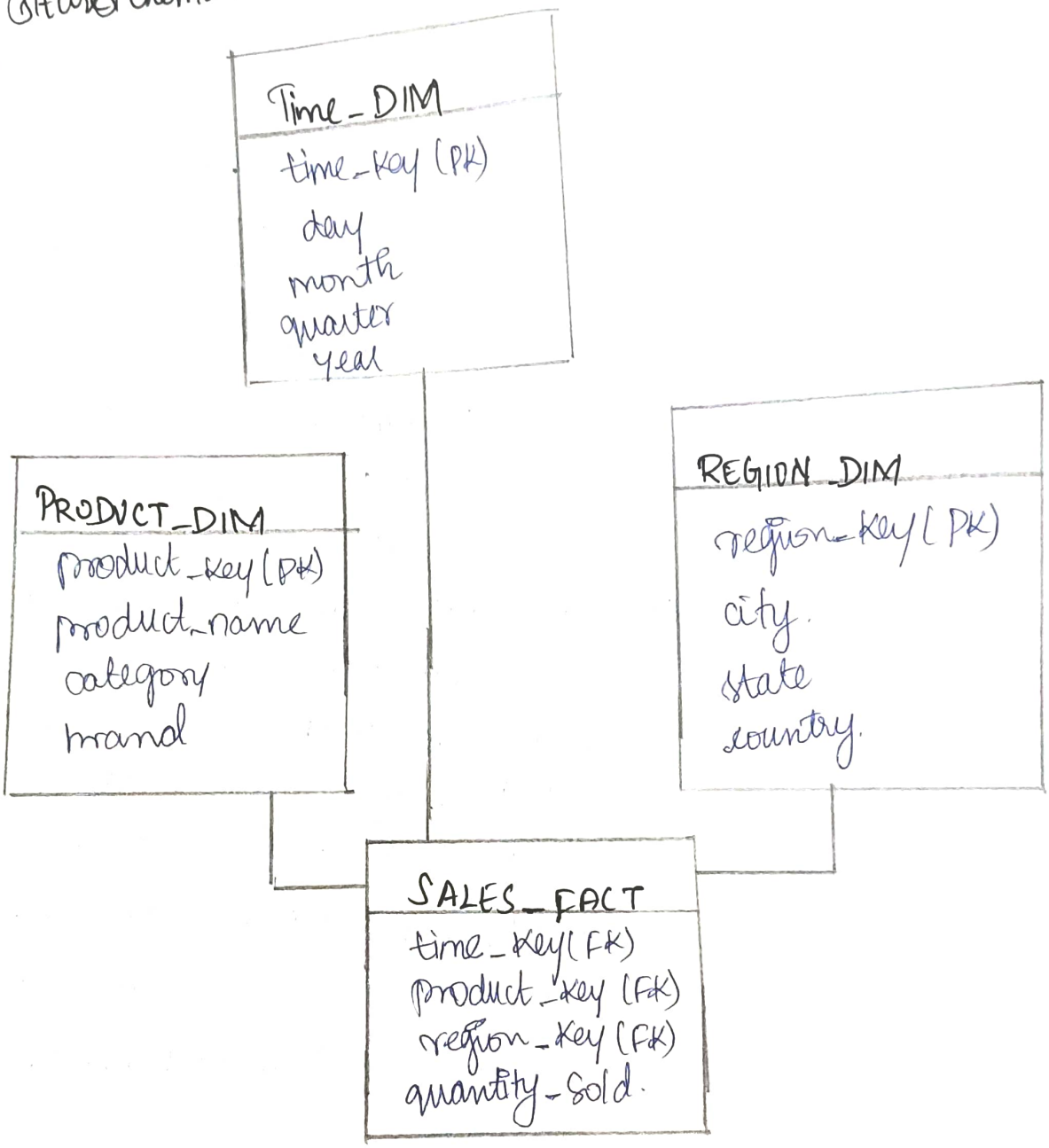
Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30%**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25%**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20%**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15%**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10%**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Question : 1 – E-commerce Data Warehouse.

a] Three commonly used Schemas:

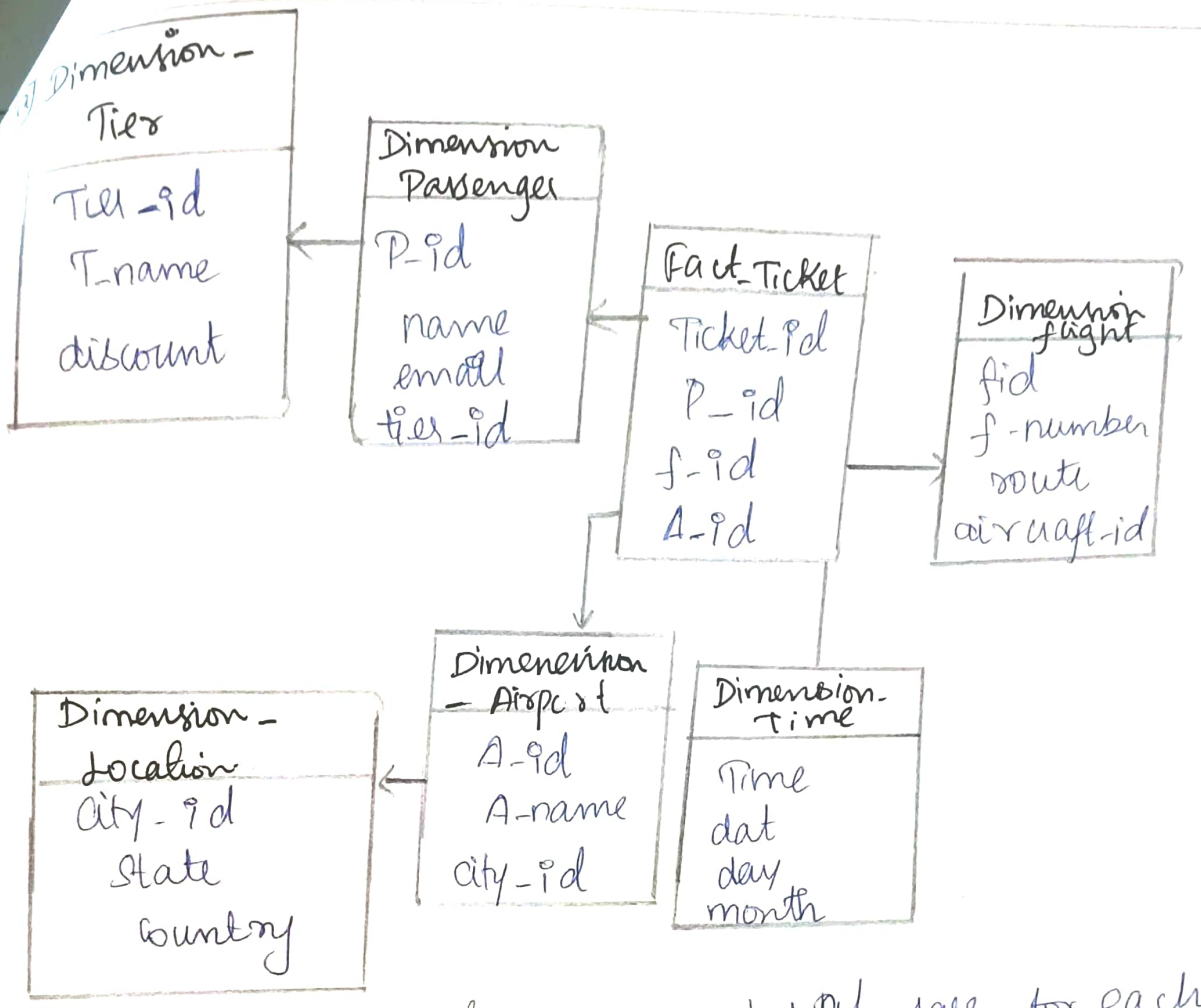
Schema	Description
Star Schema	One central fact table linked directly to denormalized dimension tables. It is simple, needs fewer joins and gives fast queries, but has some redundancy.
Snowflake schema	A variant of the star schema in which dimension tables are normalized into sub-dimensions. It saves space and reduce redundancy, but needs more joins and is slower.
Galaxy	Multiple fact tables share common dimension tables. It suits enterprise warehouse that model several business processes.

b) Star schema:



c) SQL Query:

```
SELECT region, year, month, SUM(revenue) AS monthly_revenue FROM Sales GROUP BY region, year, month ORDER BY region, year, month.
```



b) OLAP operation to find year wise total fare for each airport.

1. Rollup : Aggregate time from day/month to year.
2. Slice / Dice : Select the required airport.
3. Aggregate : calculate sum (total_fare) grouped by airport and year.

c) total no. of. cuboids:

for n dimension, total cuboids = 2^n

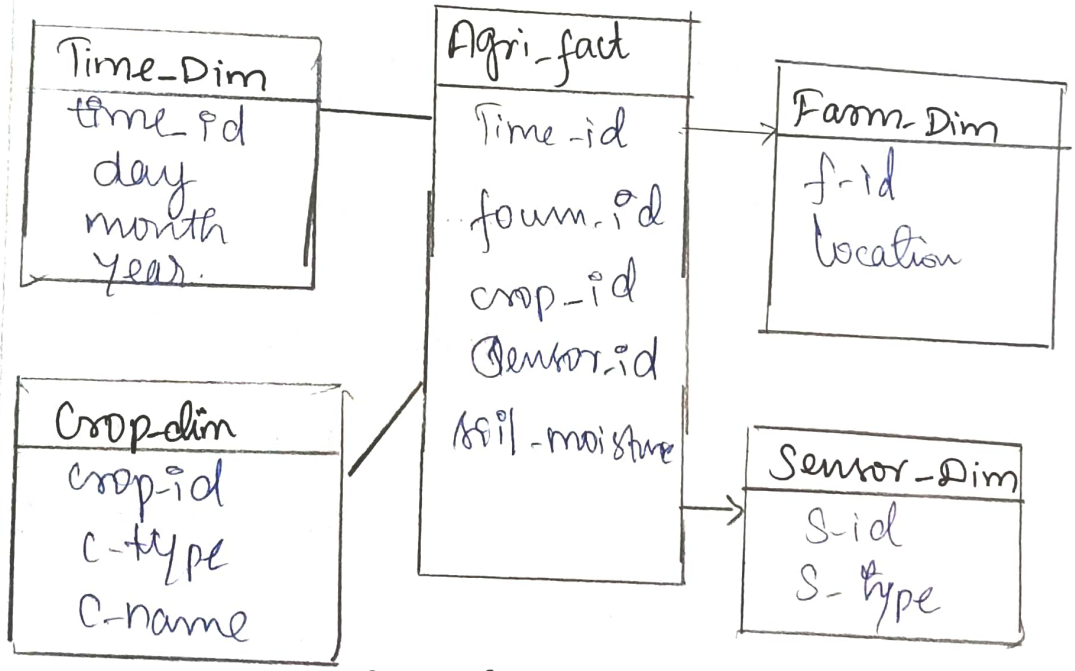
Here $n = 4$,

so total cuboids = $2^4 = 16$ cuboids

3] a] Fact and dimension tables:

- Fact table: CROP_SENSOR_FACT.
  - Foreign key: time, farm_key, crop key, sensor key.
  - Measure: soil_moisture, temperature.
- Dimension table:
  - Time: time_key, week, month, season.
  - Crop: crop_key, crop_name, cope_type.
  - Soil: soil_key, Soil_type, PH_level.

b) Star Schema:



c] Multi dimensional Analysis:

The star schema combines sensor readings, crop formation, farm information and time, OLAP operations such as roll-up, drill down, slice and dice help to analyse the crop field across different years, farms and crops.

Given:

n- dimension, each having m distinct values.

without concept hierarchies.

a) Maximum numbers of cells in the base cuboid,

Each dimension has m values, m^n

b) Minimum no. of cells in base cuboid.

the min. occurs when each dimensions has only one distinct values = 1

c) Maximum no. of all cells in online cube.

These are 2^n cuboids, and each dimensions can contribute either a specific value 2^n an Aggregated values

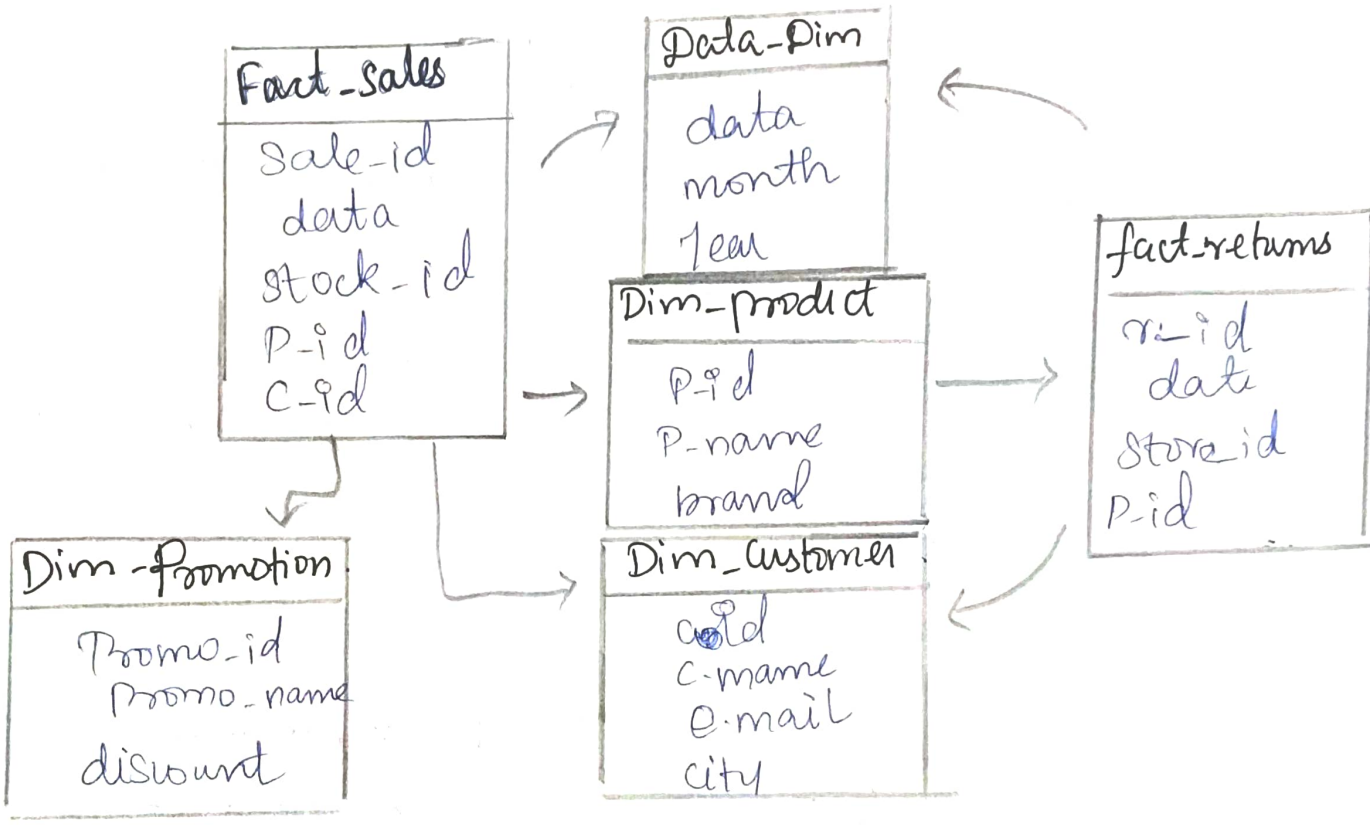
The maximum total no. of cells is $= (m+1)^n$

d) Conditions for minimum no. of cells

The cube has minimum no. of cells when each dimensions contain only one distinct values.

∴ the basic cuboid has one cell, and all aggregate cuboids also contain 1 cell each.

Question: 5] Galaxy Schema:



b] OLAP operations:

1. Rollup: Group daily records by year
2. Slice/Dice: Select required stores
3. Aggregation: Calculate total sales and total
4. Calculation: Net Sales = Salesamount - Refund amount.

c] Total no. of. cuboids

dimensions - 4, levels = 3

Standard formula for dimensions with 3 hierarchy levels: $(3+1)^4 = 4^4 = 256$ cuboids